Git & GitHub Commands Cheat Sheet

Installing Git Bash (Git for Windows)

Git Bash gives you a Linux-like terminal on Windows with Git tools.

Installation Steps (Windows):

1. Visit → https://git-scm.com/download/win
2. Download and run the `.exe` installer.
3. Follow the wizard with these recommended settings:
    - Select **Git Bash Here** integration
    - Add Git to **PATH**
    - Choose **VS Code** as default editor
    - Use **MinTTY** terminal
    - Use Windows-style line endings conversion
4. Finish installation.

Verify Installation:

```
# Open Git Bash and run:
git --version                        # Shows installed Git version

# First-time global setup (run once)
git config --global user.name "Satyasai Esarapu"      # Set your name
git config --global user.email "your.email@example.com"   # Set your email
git config --list                    # View all config settings
```

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1. Configuration

```
git config --global user.name "Your Name"        # Sets your name for a
git config --global user.email "you@example.com"  # Sets your email for
git config --list                                 # Displays all Git cor
git config --global core.editor "code --wait"     # Sets VS Code as defa
```

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2. Repository Setup

```
git init                                         # Initialize a new loc
git clone https://github.com/user/repo.git       # Clone (download) a r
git clone git@github.com:user/repo.git           # Clone using SSH (rec
```

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3. Staging & Committing

```
git status                                       # Shows current state
git add filename.txt                             # Stage a specific fil
git add .                                        # Stage all modified &
git add -A                                       # Stage all changes (i

git reset filename.txt                           # Unstage a file (keep

git commit -m "Add login feature"                # Commit staged change
git commit                                       # Commit with default
git commit --amend -m "Fixed typo"               # Modify the last comm
```

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4. Branching & Merging

```
git branch                                       # List all local branc
git branch -a                                    # List all branches (l

git checkout -b feature-branch                   # Create AND switch to
git checkout main                                # Switch to an existir
git branch feature-branch                        # Create branch withou

git merge feature-branch                         # Merge changes from f
git branch -d feature-branch                     # Delete merged branch
git branch -D feature-branch                     # Force delete branch
```

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5. Remote & GitHub Operations

```
git remote add origin https://github.com/user/repo.git   # Link local rep
git remote -v                                            # View all remote repo

git push origin main                                     # Upload local commits
git push -u origin main                                  # Push + set upstream

git pull origin main                                     # Download and merge l
git fetch origin                                         # Download changes wit

git push origin --delete old-branch                      # Delete a remote bran
```

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6. Viewing History & Changes

```
git log                                                  # Full commit history
git log --oneline --graph --all                          # Compact graph view c
git diff                                                 # Show unstaged change
git diff filename.txt                                    # Show changes in a sp
git show abc1234                                         # Show details of a sp
git blame filename.txt                                   # Show who last change
```

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7. Undoing Changes

```
git reset                                                # Unstage all files
git restore filename.txt                                 # Discard local change
git checkout -- filename.txt                             # Same as above (older

git revert abc1234                                       # Create a new commit
git reset --soft HEAD~1                                  # Undo last commit but
git reset --hard HEAD~1                                  # Undo last commit AND
```

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8. Stashing (Temporary Save)

```
git stash                                                # Temporarily save you
git stash push -m "WIP: login UI"                        # Stash with a custom
git stash list                                           # List all stashed cha
```

```
git stash pop                                    # Apply latest stash a
git stash apply stash@{1}                        # Apply a specific sta
```

Pro Tips

- Always run `git status` before any major action
- Write clear, meaningful commit messages
- Pull before you push: `git pull --rebase origin main`
- Use `.gitignore` to avoid committing unwanted files